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Task 1 — Mountain Named Entity Recognition

Solution overview

Built a synthetic dataset from 43 real mountain names with known heights. About 30 sentence templates with randomized verbs, adjectives, and height facts, plus hard negatives — the same templates applied to rivers, lakes, cities, and people's names — so the model doesn't just tag any capitalized word as a mountain. 581 sentences total, BIO-labelled, split 80/10/10.

Fine-tuned bert-base-cased (casing matters — mountain names are always capitalized) with a token-classification head, 5 epochs, AdamW. Kept the checkpoint with the best validation F1.

Results

Test F1 = 1.00. But the test set comes from the same template generator as training, so this mostly shows the model learned the templates, not that it generalizes. A better check is probing it with sentences it never saw:

```
Kilimanjaro is the tallest freestanding mountain in Africa.
  → Kilimanjaro, conf 0.52
Denali attracts climbers every summer.
  → Denali, conf 0.73
Scientists recently studied glacier melt on Mount Logan in Canada.
  → Mount Logan, conf 0.996 (not in the training data)
Thousands of tourists visit Paris every single year.
  → nothing detected, correct
On our trip we visited both Mount Fuji and Tokyo.
  → Mount Fuji, conf 0.996, Tokyo ignored correctly
```

Good sign: it generalized to Mount Logan, a mountain it never saw, and correctly ignored Paris and Tokyo even next to a real mountain.

Bad sign: confidence drops a lot (0.52–0.73 vs. 0.99+) on single-word names at the start of a sentence, versus "Mount X" mid-sentence. Likely cause — most templates put "Mount X" mid-sentence, so the model leans on the word "Mount" and position as its main signals.

Potential improvements

- **Bigger base model.** roberta-large or deberta-v3-large usually add 3-5 F1 points on NER, no data changes needed.
- **More, harder data.** pull names from Wikidata (~15,000), add templates for single-word names at sentence start, mix in real text from Wikipedia or news.
- **Data augmentation.** back-translation, swapping mountain names between sentences, LLM paraphrasing.
- **CRF layer.** on top of BERT to enforce valid tag sequences.
- **Span-based NER.** SpanBERT or SpERT instead of per-token tagging.
- **Gazetteer feature.** a simple "is this token a known mountain name" flag added to the embeddings.
- **Keep mining hard negatives.** cases where the model confuses "Mount" as part of a person's or place's name.
- **Cross-lingual transfer.** xlm-roberta-large for other languages, no extra labelling needed.

Task 2 — Sentinel-2 Image Matching Across Seasons

Solution overview

Found Sentinel-2 TCI tiles of the same location from two seasons (winter/spring and summer) and downloaded just those, instead of the full ~35 GB dataset. Cut 20 matching 512x512 crops from the same pixel coordinates in both scenes — Sentinel-2 tiles are pixel-aligned across dates, so the same row/col range is the same spot on the ground regardless of season. Added light CLAHE contrast enhancement since raw satellite composites look flat.

Used LoFTR (via kornia) instead of SIFT/ORB or SuperPoint+SuperGlue — it matches directly from coarse-to-fine features without a separate keypoint-detection step, which handles seasonal changes (snow, vegetation, lighting) much better. Started from the indoor pretrained checkpoint and fine-tuned on satellite patches, using random homographies (rotation ±20°, scale ±15%, shift ±15%) as synthetic ground truth, so no manual labelling was needed.

Large scenes are split into 512x512 tiles with 64px overlap, matched independently, then merged; RANSAC filters out bad matches afterward.

Caveat: the fine-tuning ground truth is synthetic (homographies on single-season crops) — it teaches geometric robustness, not real cross-season appearance robustness. The dataset itself was verified end-to-end on real imagery (20 correctly-aligned winter/summer pairs); the fine-tuning numbers should be double-checked against actual training logs before reporting them as final.

Potential improvements

- **Stronger backbone.** an outdoor/aerial-pretrained checkpoint (EfficientLoFTR, DeDoDe) instead of the indoor one.
- **Multi-spectral input.** near-infrared or false-colour bands instead of just visible RGB — more stable across seasons.
- **Real correspondence data.** replace synthetic homographies with real cross-season pairs with known GPS-derived alignment.
- **Bigger patches.** 512x512 instead of 256x256 gives more context and better matches on flat areas like fields or water.
- **Smarter tile merging.** confidence-weighted merging instead of simple proximity dedup, to reduce duplicates at tile borders.
- **Better outlier rejection.** MAGSAC++ or a learned method (OANet) instead of plain RANSAC, especially for repetitive scenes like farmland.
- **Radiometric normalization.** histogram matching between image pairs before inference to close the seasonal appearance gap.
- **Cross-sensor training.** mix in Landsat/PlanetScope pairs so the model isn't locked to Sentinel-2 only.